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DRUG REPURPOSING DEMO
Query   : 'alzheimer's'  →  Alzheimer disease
Model   : HGT
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Scoring 100 drug candidates ...
Computing disease similarity ...
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OUTPUT 1 — TOP 5 CANDIDATE DRUGS
Disease : Alzheimer disease  |  Model : HGT
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Rank	Drug	Model Score	Sim. Support	Rerank Score	Risk	Evidence	Safety Flags	Known Treatment For
1	Istradefylline (drug_5977)	0.568464	0.237003	0.571906	low	6	0	X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson dise...
2	Safinamide (drug_3813)	0.568407	0.237003	0.571875	low	7	0	X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson dise...
3	Droxidopa (drug_3666)	0.568343	0.237003	0.571839	low	5	0	X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson dise...
4	Bromocriptine (drug_1162)	0.568243	0.237003	0.571784	low	5	0	X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson dise...
5	Lithium carbonate (drug_6544)	0.568421	0.227618	0.569536	low	4	0	headache disorder, manic bipolar affective disorder, juvenile arthritis due to defect in LACC1 (+5 more)

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OUTPUT 2 — TOP 10 SIMILAR DISEASES
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Rank	Similar Disease	Similarity Score	Embedding Cos	Shared Proteins	Shared Effects	Shared Drugs
1	Alzheimer disease without neurofibrillary tangles (disease_1207)	0.777324	0.980951	101	0	2
2	dementia (disease) (disease_4029)	0.660359	0.951862	6	0	0
3	Pick disease (disease_15540)	0.625575	0.975548	101	3	3
4	Parkinson disease (disease_1428)	0.621532	0.977789	18	17	0
5	cerebellar degeneration (disease_8600)	0.620203	0.973985	101	0	4
6	hereditary breast carcinoma (disease_4152)	0.610758	0.952930	23	2	0
7	Reye syndrome (disease_13495)	0.599356	0.996321	101	0	5
8	anxiety disorder (disease_15492)	0.597925	0.964913	29	1	0
9	major affective disorder (disease_14864)	0.596919	0.962324	23	2	0
10	familial prostate carcinoma (disease_8836)	0.596704	0.986251	22	1	0

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OUTPUT 3 — EVIDENCE & SIMILARITY SUBGRAPH
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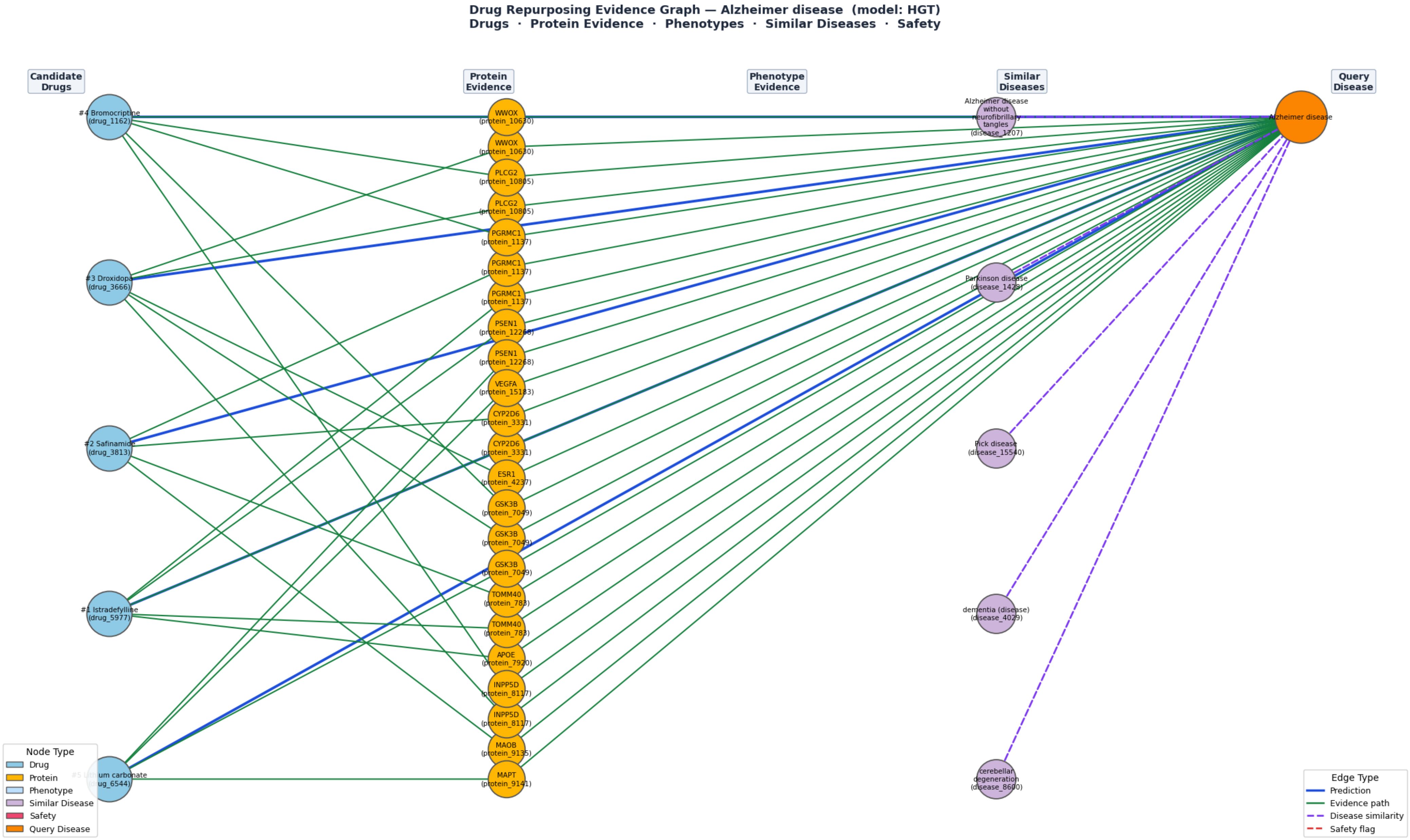

Subgraph saved → /mnt/data/demo_subgraph.png

OUTPUT 4 — RAG EXPLANATION & SUMMARY

DRUG REPURPOSING REPORT

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Disease query   : Alzheimer disease
Model           : HGT

SIMILAR DISEASES USED AS CONTEXT
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Diseases most biologically similar to the query were identified using:
embedding cosine similarity · disease ontology edges · shared proteins
PPI-expanded protein bridges · phenotype overlap · shared therapeutics

  • Alzheimer disease without neurofibrillary tangles (disease_1207) (similarity=0.777, embedding=0.981, shared proteins=101, shared effects=0)
  • dementia (disease) (disease_4029) (similarity=0.660, embedding=0.952, shared proteins=6, shared effects=0)
  • Pick disease (disease_15540) (similarity=0.626, embedding=0.976, shared proteins=101, shared effects=3)
  • Parkinson disease (disease_1428) (similarity=0.622, embedding=0.978, shared proteins=18, shared effects=17)
  • cerebellar degeneration (disease_8600) (similarity=0.620, embedding=0.974, shared proteins=101, shared effects=0)

TOP 5 CANDIDATE DRUGS
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Rank 1 — Istradefylline (drug_5977)
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Currently known to treat : X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson disease (+5 more)
Raw model score          : 0.568
Similar disease support  : 0.237
Final rerank score       : 0.572
Risk assessment          : low observed graph-based safety concern

Mechanistic evidence:
  • Shared protein target: Istradefylline (drug_5977) --[drug_enzyme_protein]--> CYP2D6 (protein_3331) <--[disease_associated_with_protein]-- Alzheimer disease (disease_1510)
  • Protein interaction bridge: Istradefylline (drug_5977) --[drug_carrier_protein]--> ALB (protein_4264) --[protein_interacts_with_protein]--> APOE (protein_7920) <--[disease_associated_with
  • Protein interaction bridge: Istradefylline (drug_5977) --[drug_enzyme_protein]--> CYP1A2 (protein_3296) --[protein_interacts_with_protein]--> PGRMC1 (protein_1137) <--[disease_associated_
  • Protein interaction bridge: Istradefylline (drug_5977) --[drug_enzyme_protein]--> CYP2C8 (protein_3327) --[protein_interacts_with_protein]--> PGRMC1 (protein_1137) <--[disease_associated_

Evidence profile:
  • Shared protein bridges         : 1
  • Protein interaction bridges    : 5
  • Disease-to-disease transfer    : 0
  • Phenotype / effect overlap     : 0

Safety signals:
  • No direct contraindication or adverse overlap signal found.

Support from similar diseases:
  Parkinson disease (disease_1428)
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Rank 2 — Safinamide (drug_3813)
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Currently known to treat : X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson disease (+4 more)
Raw model score          : 0.568
Similar disease support  : 0.237
Final rerank score       : 0.572
Risk assessment          : low observed graph-based safety concern

Mechanistic evidence:
  • Shared protein target: Safinamide (drug_3813) --[drug_target_protein]--> MAOB (protein_9135) <--[disease_associated_with_protein]-- Alzheimer disease (disease_1510)
  • Shared protein target: Safinamide (drug_3813) --[drug_enzyme_protein]--> CYP2D6 (protein_3331) <--[disease_associated_with_protein]-- Alzheimer disease (disease_1510)
  • Protein interaction bridge: Safinamide (drug_3813) --[drug_enzyme_protein]--> CYP1A2 (protein_3296) --[protein_interacts_with_protein]--> PGRMC1 (protein_1137) <--[disease_associated_with
  • Protein interaction bridge: Safinamide (drug_3813) --[drug_enzyme_protein]--> CYP2C9 (protein_3328) --[protein_interacts_with_protein]--> TOMM40 (protein_783) <--[disease_associated_with_

Evidence profile:
  • Shared protein bridges         : 2
  • Protein interaction bridges    : 5
  • Disease-to-disease transfer    : 0
  • Phenotype / effect overlap     : 0
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Safety signals:
  • No direct contraindication or adverse overlap signal found.

Support from similar diseases:
  Parkinson disease (disease_1428)
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Rank 3 — Droxidopa (drug_3666)
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Currently known to treat : X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson disease (+4 more)
Raw model score          : 0.568
Similar disease support  : 0.237
Final rerank score       : 0.572
Risk assessment          : low observed graph-based safety concern

Mechanistic evidence:
  • Protein interaction bridge: Droxidopa (drug_3666) --[drug_target_protein]--> ADRA2A (protein_3097) --[protein_interacts_with_protein]--> GSK3B (protein_7049) <--[disease_associated_with_p
  • Protein interaction bridge: Droxidopa (drug_3666) --[drug_target_protein]--> ADRA2A (protein_3097) --[protein_interacts_with_protein]--> INPP5D (protein_8117) <--[disease_associated_with_
  • Protein interaction bridge: Droxidopa (drug_3666) --[drug_target_protein]--> ADRA2A (protein_3097) --[protein_interacts_with_protein]--> PLCG2 (protein_10805) <--[disease_associated_with_
  • Protein interaction bridge: Droxidopa (drug_3666) --[drug_target_protein]--> ADRA2C (protein_3203) --[protein_interacts_with_protein]--> WWOX (protein_10630) <--[disease_associated_with_p

Evidence profile:
  • Shared protein bridges         : 0
  • Protein interaction bridges    : 5
  • Disease-to-disease transfer    : 0
  • Phenotype / effect overlap     : 0

Safety signals:
  • No direct contraindication or adverse overlap signal found.

Support from similar diseases:
  Parkinson disease (disease_1428)
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Rank 4 — Bromocriptine (drug_1162)
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Currently known to treat : X-linked parkinsonism-spasticity syndrome, early-onset parkinsonism-intellectual disability syndrome, Parkinson disease (+7 more)
Raw model score          : 0.568
Similar disease support  : 0.237
Final rerank score       : 0.572
Risk assessment          : low observed graph-based safety concern

Mechanistic evidence:
  • Protein interaction bridge: Bromocriptine (drug_1162) --[drug_enzyme_protein]--> CYP3A4 (protein_3344) --[protein_interacts_with_protein]--> PGRMC1 (protein_1137) <--[disease_associated_w
  • Protein interaction bridge: Bromocriptine (drug_1162) --[drug_target_protein]--> ADRA2A (protein_3097) --[protein_interacts_with_protein]--> GSK3B (protein_7049) <--[disease_associated_wi
  • Protein interaction bridge: Bromocriptine (drug_1162) --[drug_target_protein]--> ADRA2A (protein_3097) --[protein_interacts_with_protein]--> INPP5D (protein_8117) <--[disease_associated_w
  • Protein interaction bridge: Bromocriptine (drug_1162) --[drug_target_protein]--> ADRA2A (protein_3097) --[protein_interacts_with_protein]--> PLCG2 (protein_10805) <--[disease_associated_w

Evidence profile:
  • Shared protein bridges         : 0
  • Protein interaction bridges    : 5
  • Disease-to-disease transfer    : 0
  • Phenotype / effect overlap     : 0

Safety signals:
  • No direct contraindication or adverse overlap signal found.

Support from similar diseases:
  Parkinson disease (disease_1428)
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Rank 5 — Lithium carbonate (drug_6544)
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Currently known to treat : headache disorder, manic bipolar affective disorder, juvenile arthritis due to defect in LACC1 (+5 more)
Raw model score          : 0.568
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Similar disease support  : 0.228
Final rerank score       : 0.570
Risk assessment          : low observed graph-based safety concern

Mechanistic evidence:
  • Shared protein target: Lithium carbonate (drug_6544) --[drug_target_protein]--> GSK3B (protein_7049) <--[disease_associated_with_protein]-- Alzheimer disease (disease_1510)
  • Protein interaction bridge: Lithium carbonate (drug_6544) --[drug_target_protein]--> GSK3B (protein_7049) --[protein_interacts_with_protein]--> MAPT (protein_9141) <--[disease_associated_
  • Protein interaction bridge: Lithium carbonate (drug_6544) --[drug_target_protein]--> GSK3B (protein_7049) --[protein_interacts_with_protein]--> PSEN1 (protein_12268) <--[disease_associate
  • Protein interaction bridge: Lithium carbonate (drug_6544) --[drug_target_protein]--> GSK3B (protein_7049) --[protein_interacts_with_protein]--> VEGFA (protein_15183) <--[disease_associate

Evidence profile:
  • Shared protein bridges         : 1
  • Protein interaction bridges    : 3
  • Disease-to-disease transfer    : 0
  • Phenotype / effect overlap     : 0

Safety signals:
  • No direct contraindication or adverse overlap signal found.

Support from similar diseases:
  major affective disorder (disease_14864)

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INTERPRETATION NOTE
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These are graph-based research hypotheses, not clinical prescriptions.
The strongest candidates combine: high model score + strong mechanistic
evidence (especially shared protein bridges) + support from biologically
similar diseases + no direct contraindication signals.

Always validate predictions against current clinical literature before
drawing any experimental or clinical conclusions.
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  Structured RAG report table:
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